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Federated Deep Learning for Scalable and Privacy-Preserving Distributed Denial-of-Service Attack Detection in Internet of Things Networks

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Abstract: Industry-wide IoT networks have altered operations and increased vulnerabilities, notably DDoS attacks. IoT systems are decentralised. Therefore, these attacks flood networks with malicious traffic, creating interruptions, financial losses, and availability issues. We need scalable, privacy-preserving, and resource-efficient IoT intrusion detection algorithms to solve this essential problem. This paper presents a Federated-Learning (FL) framework using ResVGG-SwinNet, a hybrid deep-learning architecture, for multi-label DDoS attack detection. ResNet improves feature extraction, VGGNet optimises feature refining, and Swin-Transformer captures contextual dependencies, making the model sensitive to complicated attack patterns across varied network circumstances. Using the FL framework, decentralised training protects data privacy and scales and adapts across diverse IoT contexts. New preprocessing methods like Dynamic Proportional Class Adjustment (DPCA) and Dual Adaptive Selector (DAS) for feature optimisation improve system efficiency and accuracy. The model performed well on CIC-DDoS2019, UNSW-NB15, and IoT23 datasets, with 99.0% accuracy, 2.5% false alert rate, and 99.3% AUC. With a 93.0% optimisation efficiency score, the system balances computational needs with robust detection. With advanced deep-learning models, FL provides a scalable, safe, and effective DDoS detection solution that overcomes significant shortcomings in current systems. The framework protects IoT networks from growing cyber threats and provides a complete approach for current IoT-driven ecosystems.

Keywords: IoT security; DDoS detection; federated learning; deep-learning architecture; ResVGG-SwinNet; intrusion detection system

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1. Introduction

Digital transformation has begun with the rapid rise of networked devices and smart systems across industries. IoT has become a cornerstone technology, altering how devices connect and perform in complicated ecosystems [1]. Healthcare, industrial automation,

and smart cities use IoT devices for real-time data sharing and communication [2]. Modern technologies like AI, Big Data, and Cloud Computing improve IoT systems' analytical and operational capabilities, accelerating this shift [3]. Even with these advances, IoT networks' interconnectedness and pervasiveness make them vulnerable to DDoS attacks [4]. IoT networks are linked. Therefore, DDoS attacks flood targeted systems with malicious traffic, making them unavailable to legitimate users [5]. Such attacks interrupt services, cost money, and damage user confidence. IoT devices' decentralization and weak security worsen these vulnerabilities. Cheapness typically trumps security in IoT devices. They often have inadequate encryption and protection systems, making them ideal targets for attackers [6,7]. The variability and dynamic nature of IoT networks make security framework standardization difficult, stressing the necessity for specific solutions.

Federated Learning (FL) provides a possible solution to IoT security issues. FL lets devices train independently on their data and share only model updates with a central server, enabling distributed learning [8]. Decentralized systems save computing costs and improve data privacy by not transmitting raw data. FL's adaptive and scalable real-time intrusion detection and mitigation framework fits IoT networks' decentralized design. However, FL-based IDSs' efficacy depends on their capacity to handle numerous vital aspects. IoT devices create massive data streams, network traffic is dynamic, and emerging attack patterns must be detected quickly. Traditional IDSs rely on centralized designs, prone to single points of failure, and lack scalability [9]. Due to the transmission and storage of sensitive data in centralized systems, user privacy might be compromised. These constraints highlight the necessity for improved, FL-enabled IDSs that can address IoT security issues.

FL with a hybrid deep-learning architecture increases IoT DDoS attack detection scalability, efficiency, and privacy. The approach captures complicated network traffic patterns using CNNs' robust feature extraction and transformer-based models' flexibility. FL trains the model using decentralized data across several devices, protecting user privacy and efficiency. Decentralization reduces data centralization concerns and enhances system response time, which diminishes DDoS attacks [10]. Integrating FL with sophisticated deep-learning algorithms fills various literature gaps. Many DDoS detection systems cannot adapt to IoT traffic's heterogeneity. Many feature selection approaches ignore network data contextual dependencies. The proposed architecture addresses these issues using a federated method to improve detection accuracy, false positives, and scalability across massive IoT networks [11]. For high-traffic situations, hybrid optimization techniques in traffic routing improve system resilience to congestion and delays, assuring ongoing service delivery.

The framework presents a federated deep-learning strategy for detecting scalable and privacy-preserving IoT DDoS attacks. The system detects complicated attack patterns with minimal resource use using a hybrid model incorporating ResNet for feature extraction, GhostNet for computational efficiency, and Swin-Transformer for contextual dependencies. Federated learning avoids centralized data aggregation, protecting user privacy and reducing transmission overhead. Additionally, dynamic optimization approaches improve the framework's adaptability to diverse and developing IoT contexts. This complete method solves intrusion detection system constraints and lays the groundwork for secure IoT network management, assuring high scalability, flexibility, and operational integrity against growing cyber threats. The following are the main contributions of this work:

1. The ResVGG-SwinNet model has been designed, which combines ResNet, VGGNet, and Swin-Transformer. Hybrid federated learning guarantees privacy while attaining outstanding accuracy in DDoS detection.

2. The Dual Adaptive Selector (DAS) for relevance-based feature selection and redundancy reduction has been designed, thereby greatly improving computational efficiency and model accuracy.
3. Novel Preprocessing Methods: To properly handle data heterogeneity and class imbalance, Adaptive Feature Normalising (AFN), Contextual Traffic Augmentation (CTA), and Multi-scale Temporal Balancing (MTB) are used.
4. Multi-label Sensitivity Index (MSI) and Optimisation Efficiency Score (OES) have been designed for a more complex and all-encompassing multi-label DDoS detection capability assessment.
5. A federated-learning-based intrusion detection system has been designed, ensuring strong DDoS detection in over-dispersed IoT networks while respecting user data privacy and adjusting to changing network traffic circumstances.

Structure of the article: Section 2 evaluates relevant literature to identify gaps in IoT-based intrusion detection and federated learning for DDoS mitigation. Section 3 discusses the proposed system's hybrid ResVGG-SwinNet architecture, preprocessing, feature optimisation, and federated-learning framework. Section 4 shows simulation data, performance metrics, comparison analysis, and statistical assessments to prove the model works. Section 5 closes the study and suggests further research to improve the framework's scalability and usability in varied IoT scenarios.

2. Related Work

The fast spread of IoT devices and networks has improved healthcare, industrial automation, and smart cities. However, these advances present significant cybersecurity difficulties, notably in fighting against DDoS attacks. Federated-Learning (FL), Machine-Learning (ML), and hybrid deep-learning models have been used in many research studies to solve these problems. This section discusses 16 influential field publications, including their goals, methods, results, and limitations.

An FL-LSTM technique was presented to identify cyberattacks in the Internet of Health Things (IoHT) [12]. The solution used embedded feature selection to reduce computational complexity and achieved 97.16% accuracy on real-world IoHT data. The research only examined LSTM networks, which may restrict their application to various IoT scenarios with varied data patterns. Another study emphasized real-time detection and privacy preservation in IoHT security using AI [13]. This work was informative but did not evaluate hybrid systems for various attack scenarios. Another study developed an FL-based IDS for IoT networks, employing differential privacy to improve security during gradient transfers [14]. The algorithm protected data privacy but struggled to identify sophisticated attack patterns due to shallow learning. An improved Probabilistic Hybrid Ensemble Model (PHEC) for decentralized IoT networks showed label noise robustness but lacked scalability for large-scale deployments [15].

We created ML-based solutions for UAV eavesdropping and GPS spoofing attacks, obtaining excellent detection accuracy [16]. However, single-network models like LSTMs were less adaptable to changing attack patterns. Hybrid models combining LSTMs and GRUs were proposed to detect anomalies in UAV networks [17]. These methods showed that dynamic IoT traffic requires more scalable and flexible solutions. To identify zero-day botnet attacks in IoT contexts, Federated Deep Learning (FDL) was created [18]. High classification accuracy and low communication overhead were obtained; however, the absence of hybrid optimization approaches hindered its resilience against complicated attack scenarios. Another research introduced FLAD, an adaptable FL model for dynamic attack profiles [19]. FLAD corrected unbalanced datasets but needed further validation in resource-constrained IoT.

An FL-based NIDS for Satellite–Terrestrial Integrated Networks (STINs) optimized resource allocation and CPU usage [20]. The model's scalability to additional IoT applications is intriguing yet untested. Using FL for anomaly detection in IoT traffic, FeDIoT showed considerable accuracy gains [21]. Its relevance to real-time settings with changing traffic patterns was not explored. A collaborative FL architecture was suggested for training IDSs across enterprises while protecting data privacy [22]. This model performed well in classification but struggled with diverse data. FedeX, an FL-based anomaly detection system for industrial control systems, demonstrated high detection rates and efficient deployment on edge devices [23]. FedeX was successful. However, its dataset dependence restricted its applicability to IoT situations.

Federated Sequence Learning (FSL) addresses IIoT time-series modelling difficulties while emphasizing data security [24]. Data heterogeneity was handled well; however, training needed a significant amount of computing power. Another research introduced FedAGRU, an FL-based model that uses attention processes for wireless intrusion detection, resulting in improved communication efficiency [25]. Its network setup dependence limited its application to varied IoT scenarios. FLAD aims to adapt to new attack profiles dynamically, delivering high accuracy and quick convergence [26]. It was too computationally intensive for resource-constrained IoT devices. Ref. [27] emphasized the benefits of decentralized FL-based systems over centralized ML and DL techniques for privacy protection. This study stressed the necessity for hybrid models with optimization strategies to improve detection.

The summarized review in Table 1 emphasizes the importance of FL and advanced deep learning in IoT security. This research provides the groundwork for scalable, privacy-preserving, and efficient intrusion detection systems for IoT-based DDoS attacks.

Table 1. Literature review summary.

Ref.	Methodology	Key Contributions	Challenges Identified
[12]	FL-LSTM	Achieved 97.16% accuracy in detecting cyberattacks in the IoHT with embedded feature selection minimizing computational complexity.	Focused solely on LSTM, limiting applicability to diverse IoT environments.
[13]	AI-driven approach for IoHT security	Enhanced real-time detection and privacy preservation in IoHT networks.	Lacked evaluation of hybrid architectures for diverse attack scenarios.
[14]	FL-based IDS with differential privacy	Preserved data privacy during gradient exchanges; effective for Industrial IoT.	Relied on shallow learning techniques, limiting detection of complex attack patterns.
[15]	Probabilistic Hybrid Ensemble Model (PHEC)	Demonstrated resilience to label noise and effective intrusion detection.	Lacked scalability for large-scale deployments.
[16]	ML-based solution for UAV security	Achieved high detection accuracy for eavesdropping and GPS spoofing attacks.	Depended on single-network models, limiting adaptability to evolving attack patterns.
[17]	Hybrid LSTM-GRU models	Enhanced anomaly detection in UAV networks.	More scalable solutions for dynamic IoT traffic are required.
[18]	Federated Deep Learning (FDL)	Achieved high accuracy and minimized communication overhead for zero-day botnet detection.	Did not integrate hybrid optimization methods, limiting robustness.
[19]	FLAD (Adaptive FL for DDoS detection)	Addressed imbalanced datasets; improved dynamic attack profile detection.	Required further validation in resource-constrained environments.

Table 1. *Cont.*

Ref.	Methodology	Key Contributions	Challenges Identified
[20]	Distributed NIDS for STINs using FL	Optimized resource allocation and improved CPU utilization in satellite–terrestrial networks.	Scalability to other IoT applications was unexplored.
[21]	FeDIoT platform with FL and autoencoders	Achieved significant accuracy improvements for anomaly detection in IoT traffic.	Did not address real-time scenarios with dynamic traffic patterns.
[22]	Collaborative FL framework	Preserved data privacy while training IDS across organizations; high classification performance.	Faced challenges in handling heterogeneous data sources.
[23]	FedeX: FL-based explainable anomaly detection	Achieved high detection rates and efficient deployment on edge devices.	Relied on specific datasets, limiting generalizability.
[24]	Federated Sequence Learning (FSL)	Effectively handled data heterogeneity in IIoT while ensuring data security.	Required extensive computational resources for training.
[25]	FedAGRU with attention mechanisms	Improved communication efficiency and accuracy for intrusion detection.	Dependency on specific network configurations restricted broader applicability.
[26]	FLAD (Adaptive FL for new attack profiles)	Rapid convergence and high accuracy for dynamic attack adaptation.	Required substantial computational resources, limiting suitability for resource-constrained devices.
[27]	Decentralized FL-based systems	Emphasized privacy preservation and enhanced detection capabilities.	Did not integrate hybrid optimization algorithms, limiting detection robustness.

3. Proposed Methodology

A scalable and reliable framework for multi-label DDoS attack detection in IoT networks is provided via a hybrid architecture of deep learning and federated learning. The innovative ResVGG-SwinNet architecture integrates ResNet, VGGNet, and Swin-Transformer to handle varied Internet of Things (IoT) traffic and ever-changing cyberattack patterns. The dataset is improved and class imbalance is reduced using advanced preprocessing approaches such as Multi-scale Temporal Balancing (MTB) and Adaptive Feature Normalisation (AFN), which in turn makes the framework efficient and adaptive. By reducing communication overhead and enabling decentralised training across several IoT devices, the federated-learning paradigm effectively updates models while safeguarding data privacy. By reducing duplication while maintaining critical data, the Dual Adaptive Selector (DAS) improves the predictive value of features via the application of unique feature optimization algorithms. To further enhance the model's ability to identify multi-label attacks, techniques for feature generation and calibration are used. These algorithms capture intricate relationships and temporal dynamics within the dataset. By centrally updating client-specific models and then aggregating them, federated learning trains the ResVGG-SwinNet architecture to produce a fast, privacy-preserving global model. We integrate all parts to provide adaptability to different IoT settings, low false positive rates, and excellent detection accuracy. These aspects are described in each of the sections that follow. There is a visual representation of the suggested system framework in Figure 1.

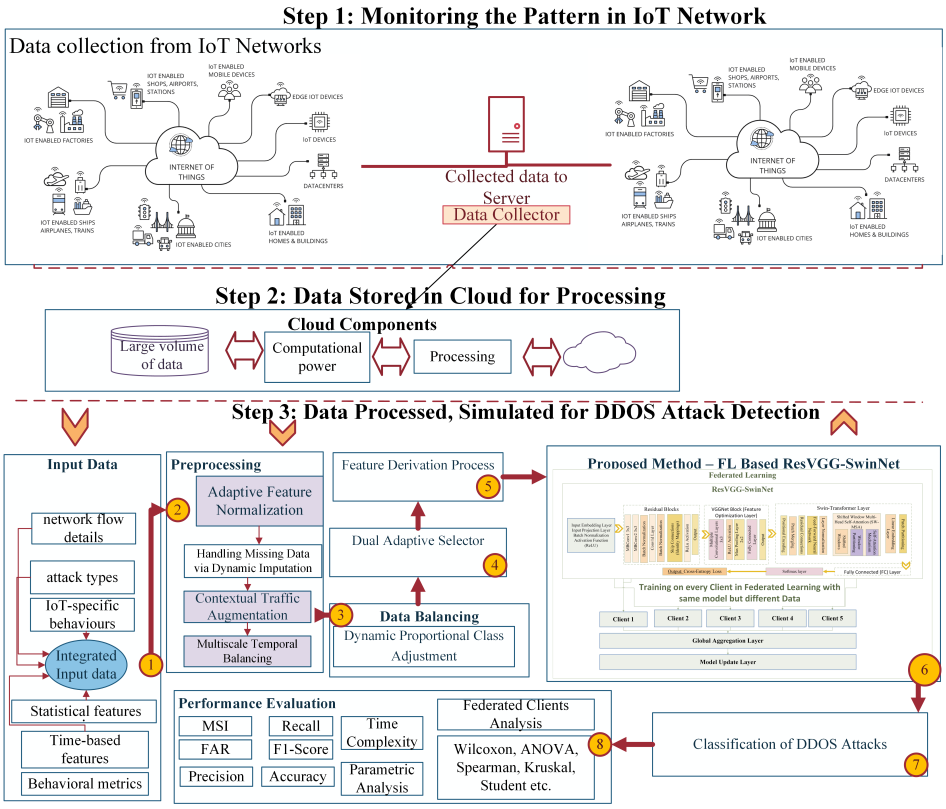


Figure 1. Proposed FL-based DDOS attack detection framework.

3.1. Dataset Description

This research used three datasets to construct a federated deep-learning system for scalable and privacy-preserving DDoS assault detection. These network traffic statistics include benign and DDoS assault situations and are commonly used in cybersecurity research. The IoT23 dataset records network traffic from IoT devices, covering both benign behaviour and threats [28]. This dataset is useful for studying IoT network traffic under normal and malicious settings. UNSW-NB15 [29] and CIC-DDOS2019 [30] are benchmark datasets that simulate current network traffic, including several attack methods, making them useful for analysing statistical and behavioural metrics. The IoT23 Network Attack Dataset focuses on real-time traffic analysis to identify abnormalities like DDoS assaults and offers a diversity of network traffic patterns. Multi-label DDoS detection using these datasets may identify many assault types in distributed systems. Table 2 provides feature explanations for each dataset. The number in the step 3 of figure shows the work flow sequence of analyzing the data.

Table 2. Dataset features overview.

SNo.	IoT23 Features	UNSW-NB15 Features	IoT23 Network Attack Dataset Features
1	- Includes network flow details such as source and destination IPs, ports, and protocols. - Covers attack types such as Mirai botnet, DDoS, and benign IoT traffic patterns. - Offers insights into IoT-specific behaviours, both benign and malicious.	- Statistical features representing traffic metrics like packet count, byte count, and flow duration. - Time-based features like start time, end time, and connection duration. - Behavioral metrics, including TCP flags, retransmissions, and malformed packets.	- Packet-level details capturing anomalies in network traffic, including timestamps, packet size, and protocol. - Includes labelled data for DDoS attacks and benign traffic scenarios. - Designed for anomaly detection, with features supporting real-time traffic analysis.

3.2. Preprocessing Steps

To provide clean, consistent data for multi-label DDoS attack detection, preprocessing techniques were done after dataset preparation [31]. Novel preprocessing approaches improved network traffic representation while maintaining dataset structure. AFN, CTA, and MTB were used. Adaptive Feature Normalization (AFN) was used to scale features while dynamically accounting for feature-specific variability. Unlike standard normalization, AFN modifies the scaling range depending on feature distribution across the attack and normal traffic categories. The transformation for each feature y_j is presented in Equation (1):

$$\widetilde{y}_j = \frac{y_j - \nu_{y_j,\text{grp}}}{\eta_{y_j,\text{grp}} + \epsilon} \tag{1}$$

where $\nu_{y_j,\text{grp}}$ and $\eta_{y_j,\text{grp}}$ represent the mean and standard deviation of feature y_j within a particular group (e.g., attack or regular), whereas ϵ prevents division by zero. This technique normalises features to their group-specific behaviour, boosting the model's sensitivity to nuanced attack patterns.

The use of Contextual Traffic Augmentation (CTA) increased training data variety. Combining weighted averages of neighbouring traffic samples with temporal and contextual relationships creates additional data points. For two traffic cases, z_p and z_q, the enhanced sample z_{aug} is calculated using Equation (2):

$$z_{\text{aug}} = \beta z_p + (1-\beta) z_q, \quad \beta = \frac{e^{-\Delta_s}}{1 + e^{-\Delta_s}} \tag{2}$$

As Δ_s indicates the temporal distance between z_p and z_q, samples closer in time are more likely to impact each other. This augmentation adds realistic variations while keeping data context.

Finally, Multi-scale Temporal Balancing (MTB) addressed class imbalance, notably attack type distribution. This approach uses sample frequency over various temporal frames to determine a balancing weight, ω_c, for each class, c. Weight assignment is stated in Equation (3):

$$\omega_c = \frac{\sum_{m=1}^{M} \phi_{c,m}}{M \cdot \max_c(\phi_{c,m})} \tag{3}$$

where$\phi_{c,m}$ indicates the frequency of class c in temporal window m, whereas M represents the total number of temporal windows. MTB avoids oversampling by dynamically redistributing samples between windows to balance all classes.

Methods for preprocessing datasets make them more amenable to federated multi-label DDoS detection by improving dataset quality. It was shown that all of the transformations handled class imbalance, feature scaling, and data variety while preserving the semantic integrity of the data.

3.3. Data Balancing

The dataset's class imbalance was remedied using Dynamic Proportional Class Adjustment (DPCA). By taking data links into account, this method dynamically scales underrepresented classes. In order to provide synthetic data for underrepresented classes and to decrease overrepresented classes proportionally, DPCA uses weighted resampling [32]. The amount of oversampling or undersampling that is required in synthetic sampling is determined by a weight factor, ζ_k, that is assigned to each class, k. Equation (4) is used to compute the weight factor.

$$\zeta_k = \frac{\overline{n} - n_k}{\max(n_k, \delta)} \tag{4}$$

where $\overline{n}$ represents the average frequency across all classes, δ is a small constant used to prevent zero division, and n_k is the frequency of class k. Classes that are below $\overline{n}$ are oversampled because they receive positive weights, whereas classes that are above $\overline{n}$ are undersampled because they receive negative weights.

Oversampling minority classes is achieved by Weighted Instance Synthesis (WIS), which interpolates between data points. Two samples, v_a and v_b, belonging to the same class are used to create a synthetic sample, v_{new}, in Equation (5).

$$v_{\text{new}} = \lambda v_a + (1 - \lambda) v_b, \quad \lambda = \frac{1}{1 + e^{-\xi}} \tag{5}$$

In this equation, ξ indicates the feature distance between v_a and v_b, weighting closer samples more during interpolation.

Frequency-depending Sampling Adjustment (FBSA) decreases the sampling probability, ρ_k, of each class, k, depending on its relative frequency to control over-represented classes. Equation (6) defines sampling probability:

$$\rho_k = \frac{\overline{n}}{n_k + \delta} \tag{6}$$

Proportionately downsampling higher-frequency classes achieves balance without losing crucial information. DPCA, WIS, and FBSA represent all classes proportionately while retaining dataset variety and integrity. This method dynamically adapts to class distributions by solving multi-label DDoS attack detection problems.

3.4. Feature Optimization Using Dual Adaptive Selector

The dataset was balanced, and a new Dual Adaptive Selector (DAS) was introduced to improve feature optimisation efficiency while maintaining predictive power [33]. A hybrid framework is created by combining a Conditional Relevance Assessor (CRA) with an Iterative Dependency Filter (IDF). The chosen features are informative and non-redundant, improving multi-label DDoS attack detection performance using CRA for relevance scoring and IDF for redundancy filtering. The Conditional Relevance Assessor examines each feature's contribution based on its conditional connection with target labels. For a feature, z_q, and target class, t_r, the relevance score, $\mathcal{R}_{q,r}$, is calculated using the conditional relevance function in Equation (7):

$$\mathcal{R}_{q,r} = \frac{\text{Cov}(z_q, t_r)^2}{\text{Var}(z_q) \cdot \text{Var}(t_r)} \tag{7}$$

In this context, $\text{Cov}(z_q, t_r)$ symbolises the covariance between feature z_q and target t_r, whereas $\text{Var}(z_q)$ and $\text{Var}(t_r)$ represent their variances. Higher scores indicate target class relevance. A global relevance rating for each feature is calculated from all target label relevance scores. The Iterative Dependency Filter (IDF) uses a similarity threshold, ϑ, to remove associated features and reduce duplication among high-ranked features. The dependence coefficient, $\mathcal{C}_{q,p}$, is calculated for two features, z_q and z_p, as illustrated in Equation (8).

$$\mathcal{C}_{q,p} = \frac{\text{Cov}(z_q, z_p)}{\sqrt{\text{Var}(z_q) \cdot \text{Var}(z_p)}} \tag{8}$$

Features with lower relevance rank are excluded from selection if $\mathcal{C}_{q,p} \geq \vartheta$. The threshold ϑ adjusts dynamically depending on feature space size to maintain the least associated features.

DAS is hybrid because CRA detects essential characteristics, and IDF ensures non-redundancy. Equation (9) defines the final set of chosen features, $\mathcal{S}_f$:

$$\mathcal{S}_f = \{z_q \in \mathcal{Z} : \mathcal{R}_{q,r} > \lambda \text{ and } \mathcal{C}_{q,p} < \vartheta, \forall z_p \neq z_q\} \quad (9)$$

where $\mathcal{Z}$ represents the initial feature set, λ represents the minimal relevance threshold, and ϑ represents the dependence threshold. Features are chosen to maximise relevance and minimise redundancy.

DAS constantly adapts to the dataset, making feature optimisation efficient in settings with many input variables. The dual strategy improves model predictive power and computational efficiency for multi-label DDoS attack detection.

3.5. Feature Derivation Process

Following feature selection, a unique Feature Derivation Process (FDP) was used to develop advanced features that improve dataset representation. This stage enhances the chosen features to capture the model's hidden data linkages, interactions, and temporal dynamics. The FDP combines NDS, TEC, and JAA to provide valuable characteristics [34]. Nonlinear Dependency Synthesis finds complicated feature pair interactions. A new interaction feature, $\theta_{i,j}$, is derived from two features, p_i and p_j, using the synthesis function in Equation (10).

$$\theta_{i,j} = \log(|p_i \cdot p_j| + 1) \quad (10)$$

This modification emphasises multiplicative connections and suppresses noise to emphasise important interactions. The logarithmic component normalises excessive values. Temporal Evolution Capturer detects DDoS-related anomalies and patterns by analysing feature trends. Equation (11) calculates the temporal evolution rate, κ_k, for a feature, q_k, seen throughout a temporal frame, Ξ:

$$\kappa_k = \frac{\sum_{t=1}^{\Xi}(q_k^t - q_k^{t-1})}{\Xi - 1} \quad (11)$$

where q_k^t represents the feature value at time t, whereas κ_k measures the average change rate inside the temporal window, Ξ.

Joint Attribute Aggregator summarises many features' behaviour into a single statistic. To calculate a joint metric, $\zeta_{m,n,o}$, for features r_m, r_n, r_o, use the aggregation formula in Equation (12).

$$\zeta_{m,n,o} = \sqrt{\frac{(r_m^2 + r_n^2 + r_o^2)}{3}} \quad (12)$$

Our approach estimates the root mean square of characteristics to balance representation and preserve variability.

The FDP creates new features (Θ) via interactions, temporal evolution, and joint aggregation. The final feature set is $\mathcal{F}_{\text{final}} = \mathcal{F}_{\text{selected}} \cup \Theta$, where $\mathcal{F}_{\text{selected}}$ comprises selection process features and Θ represents newly derived attributes.

These advanced features capture complex patterns and relationships, boosting the model's multi-label DDoS attack identification. The FDP integrates crucial information into learning by gradually extending the feature set.

3.6. Feature Transformation Process

A unique **Feature Recalibration Process (FRP)** was used to refine features and optimise their contribution to the multi-label DDoS attack detection model after feature generation. Federated learning was enabled by dividing each dataset into five parts for quicker training and scalability. These subsets were shared among clients for parallel training and data privacy. This federated technique speeds up model convergence and computing, especially for massive datasets. DRA and LIS are used in the FRP. The approaches aim to improve

the relevance and effect of each characteristic by altering their scales and significance, guaranteeing proper weighting in the model [35].

Subset Creation for Federated Learning: Each dataset was divided into five subgroups for quicker training. The division distributes data among clients in federated learning for parallel processing and faster training. The dataset $\mathcal{X}$ is separated into $n = 5$ subgroups, each represented as:

$$\mathcal{X}_i = \frac{1}{5}\mathcal{X}, \quad i = 1, 2, 3, 4, 5 \tag{13}$$

To maintain diversity and balance between clients, each subset, $\mathcal{X}_i$, comprises a representative distribution of the original dataset.

Dynamic Range Adjustment (DRA): The FRP begins with DRA, which normalises features to a consistent range to solve unequal scaling across features. To calculate recalibration for a feature, ρ_m, use Equation (14):

$$\hat{\rho}_m = \frac{\rho_m - \min(\rho_m)}{\max(\rho_m) - \min(\rho_m)} \cdot (E - F) + F \tag{14}$$

In this equation, ρ_m represents the original feature values, $\min(\rho_m)$ and $\max(\rho_m)$ represent the lowest and maximum values of feature ρ_m, and F and E represent the scaling target range (e.g., $F = 0$). This transformation standardises feature values within the required range, making them similar across scales.

Local Impact Scaling (LIS): The second strategy emphasises traits highly connected with target labels. To determine the scaling factor for each feature σ_n, consider its local contribution to the target variable τ_p. Equation (15) defines the effect scaling factor, $\kappa_{n,p}$:

$$\kappa_{n,p} = \frac{\mathrm{Cov}(\sigma_n, \tau_p)}{\mathrm{Var}(\sigma_n) + \zeta} \tag{15}$$

The covariance between feature σ_n and target label τ_p is represented by $\mathrm{Cov}(\sigma_n, \tau_p)$, the variance of feature σ_n is represented by $\mathrm{Var}(\sigma_n)$, and ζ is a. Higher impact variables are scaled up, giving them more significant model influence, whereas lower impact factors are scaled down.

Finally, the final set of recalibrated features, $\mathcal{F}_r$, is created by successively applying dynamic range adjustment and local impact scaling to all features. The whole recalibrated feature set, $\mathcal{F}_{\text{final}}$, is:

$$\mathcal{F}_{\text{final}} = \{\hat{\rho}_m, \kappa_{n,p} : \forall \rho_m, \sigma_n\} \tag{16}$$

This procedure optimises recalibrated features for machine-learning models, improving multi-label DDoS attack detection. The FRP maximises feature information and model performance by modifying and scaling feature ranges by local effect. The dataset is divided into subsets for federated learning, which speeds up training and model updates without sacrificing data privacy.

3.7. Federated Classification Method with ResVGG-SwinNet

This classification framework for multi-label DDoS attack detection uses ResVGG-SwinNet, a hybrid architecture integrating ResNet [36], VGGNet [37], and Swin-Transformer [38] components. This hybrid approach uses federated learning to improve performance, scalability, and privacy while using these models. Clients may locally train the model using their data, and the aggregated model is updated on a central server without disclosing the raw data, maintaining privacy. The model architecture is shown in Figure 2.

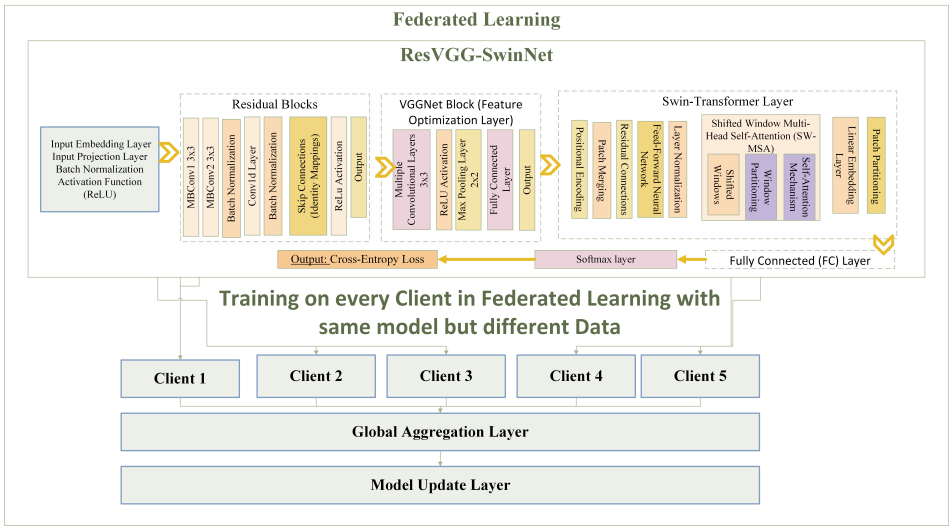


Figure 2. FL- based ResVGG-SwinNet classification model architecture.

3.7.1. Model Architecture

The ResVGG-SwinNet architecture includes ResNet, VGGNet, and Swin-Transformer. A layered model for deep-feature extraction, efficient learning, and long-term contextual comprehension uses these components in order. The ResNet Block is the primary feature-extraction layer that captures low-level features and deep representations. ResNet uses residual connections to reduce the vanishing gradient issue. This shows ResNet Block feature extraction:

$$\mathcal{F}_{\mathrm{R}}(z) = \mathcal{H}(z) + z \tag{17}$$

We use z as the input feature map and $\mathcal{H}(z)$ as the output after applying convolutional layers and residual connections. z refers to the input feature map sent through the network, whereas $\mathcal{F}_{\mathrm{R}}(z)$ holds the extracted features.

VGGNet Block (Feature Optimisation Layer): After ResNet extracts features, VGGNet optimises them. VGGNet, a deep convolutional network, employs modest (3×3) convolution filters and a basic design. After ResNet, VGGNet optimises the features:

$$\mathcal{F}_{\mathrm{V}}(z) = \phi(W \cdot \mathcal{F}_{\mathrm{R}}(z) + b) \tag{18}$$

$\mathcal{F}_{\mathrm{R}}(z)$ is the ResNet Block input, W is the weight matrix, b is the bias term, and ϕ is the activation function. Using computational efficiency, VGGNet optimises feature depth and performance.

Contextual Dependency Layer Swin-Transformer Block: The Swin-Transformer block captures long-range dependencies and contextual information for interpreting complicated data interactions. It uses window-based self-attention to learn global and local dependencies:

$$\mathcal{F}_{\mathrm{S}}(z) = \text{Swin-Transformer}(\mathcal{F}_{\mathrm{V}}(z)) \tag{19}$$

Self-attention inside non-overlapping windows and window shifts capture local and global relationships, helping the model grasp complex data patterns.

3.7.2. Federated-Learning Setup

Federated learning trains the ResVGG-SwinNet model using local datasets on each client. Federated learning protects data while distributing training. The clients train their data locally, and only the model changes (weights) are shared with a central server, which aggregates the updates and distributes the new global model to the clients.

Each client, k, trains its model on its subset of data, $\mathcal{D}_k$. The client calculates the gradient of the loss function, $\mathcal{L}$, about its data to update its local model weights, ζ_k

$$\zeta_k^{t+1} = \zeta_k^t - \gamma \nabla_{\zeta_k} \mathcal{L}(\mathcal{D}_k, \zeta_k) \tag{20}$$

ζ_k^t represents model weights at iteration t, γ represents learning rate, and $\nabla_{\zeta_k} \mathcal{L}(\mathcal{D}_k, \zeta_k)$ represents loss function gradient based on client data.

Federation Averaging (FedAvg): After local training, clients communicate their updated model weights to the central server. The central server uses Federated Averaging (FedAvg) to aggregate model changes and update the global model by averaging all client models:

$$\zeta^{t+1} = \frac{1}{N} \sum_{k=1}^{N} \zeta_k^{t+1} \tag{21}$$

where N represents client count. The aggregated model, ζ^{t+1}, is returned to the clients, and the process continues until convergence.

Global Model Update: The global model, ζ^{final}, is produced by numerous rounds of local updates and aggregations. The final model predicts and classifies, with output for each label, y_l, calculated using:

$$y_l = \text{softmax}(\mathcal{F}_S(\mathcal{F}_V(\mathcal{F}_R(z)))) \tag{22}$$

Softmax computes the probability distribution across all classes, enabling the model to categorise network traffic as usual or a particular DDoS attack.

3.7.3. Federated Learning Classification Output

After federated learning, the ResVGG-SwinNet model classifies input data as regular traffic or DDoS attacks. The model returns a probability distribution with probabilities for each class depending on its relevance to the input data using the softmax function. No raw data are exchanged across clients. Thus, federated learning offers quick model training while protecting privacy. Model aggregation and updating enhance the global model as it learns from each client's data, improving DDoS attack detection.

Federated learning and ResVGG-SwinNet provide a fast and scalable multi-label DDoS attack detection system. The model optimises deep-feature representations and learns long-range relationships utilising ResNet, VGGNet, and Swin-Transformer blocks. The federated-learning system trains the model across numerous clients while protecting data as shown in Algorithm 1.

3.8. Component-Wise Contributions of ResVGG-SwinNet

The ResVGG-SwinNet model improves multi-label DDoS detection in IoT networks by combining ResNet, VGGNet, and Swin-Transformer. These components were selected for their feature extraction, refinement, and contextual learning characteristics, enabling the model to manage IoT traffic complexity and heterogeneity.

The powerful feature extraction capabilities of ResNet, short for Residual Network, make it a popular choice. It excels in capturing packet-level statistics and flow characteristics from network traffic data. ResNet uses residual connections to avoid the vanishing gradient issue in deep networks. These connections let gradients flow through the network during backpropagation, allowing deeper layers to acquire meaningful representations without performance deterioration.

VGGNet refines features after ResNet. These modest (3×3) convolutional filters boost the depth and resolution of retrieved features in this network architecture. VGGNet refines ResNet's feature maps to help the model detect minor traffic patterns that may signal

various DDoS assaults. This modification reduces noise and improves attack detection characteristics to enhance multi-label classification.

Algorithm 1 Federated Training of ResVGG-SwinNet for DDoS Attack Detection.

Input: Local datasets $\mathcal{D}_k$ for each client, global model initialization ζ^0
Output: Final global model ζ^{final}
Initialization:
Initialize global model ζ^0 (ResNet, VGGNet, Swin-Transformer)
for each client k in parallel **do**
    Initialize local model ζ_k^0 based on global model ζ^0
    Initialize local dataset $\mathcal{D}_k$
end for
Federated-Learning Training:
for each communication round $t = 1$ to R **do**
    Local Training for Each Client
    for each client k in parallel **do**
        Perform local training on local dataset $\mathcal{D}_k$ for T iterations:
        for each iteration $i = 1$ to T **do**
            Step 1: Feature Extraction using ResNet Block
            Extract low-level features from the local data using the ResNet Block (initial feature representation).
            Step 2: Feature Optimization using VGGNet Block
            Pass the extracted features through VGGNet to optimize feature representation (adjust depth, width, and resolution).
            Step 3: Contextual Learning using Swin-Transformer Block
            Pass the optimized features through the Swin-Transformer block to capture long-range dependencies and contextual relationships.
            Step 4: Update Model Weights using Local Gradient Descent
            Compute gradients based on the loss function $\mathcal{L}$ using the local model ζ_k^i and dataset $\mathcal{D}_k$
            Update local model weights based on the computed gradients.
        end for
        Send updated local model ζ_k^{T+1} to central server
    end for
    Federated Averaging (FedAvg):
    Aggregate the models from all clients at the central server:
    $\zeta^{t+1} = \frac{1}{N} \sum_{k=1}^{N} \zeta_k^{t+1}$
    Send updated global model ζ^{t+1} back to all clients
end for
Final Model:
After R rounds, the final global model ζ^{final} is obtained
Classification:
For final global model ζ^{final}, apply softmax for multi-label classification to determine attack types (e.g., normal, DDoS).

Final component Swin-Transformer collects long-range interdependence and complicated contextual interactions in network data. The hierarchical, window-based self-attention method of the Swin-Transformer differs from local convolutional networks. Analysing traffic data at many scales lets the model understand global and local trends. This capacity helps identify changing and complex attack patterns by analysing traffic flow context and temporal relationships.

They constitute a hybrid architecture that balances feature extraction, refinement, and contextual comprehension. ResVGG-SwinNet achieves high detection accuracy, low false alarm rates, and resilient performance under different IoT network circumstances by integrating ResNet, VGGNet, and Swin-Transformer. As shown by considerable performance metrics gains, this synergy justifies the model design complexity.

4. Performance Evaluation Metrics

Performance assessment is necessary to assess ResVGG-SwinNet's multi-label DDoS attack detection performance. Traditional performance measures like accuracy, precision, recall, and F1-Score are valuable, but they do not fully address multi-label classification and DDoS detection concerns [39]. Three new measures are included to further assess the model's capacity to identify threats, minimise false positives, optimise resource utilisation, and preserve computational efficiency.

4.1. Traditional Performance Metrics

Traditional classification performance indicators like accuracy, precision, recall, and F1-Score are helpful but have drawbacks when used for multi-label classification issues. Accuracy is an overall performance statistic that ignores class imbalance and false negatives.

Precision measures the model's ability to prevent false positives or classify positive cases as true positives. The model's capacity to recognise all relevant examples is shown by the recall (or sensitivity) of true positives. The harmonic mean of accuracy and recall, the F1-Score, balances false positives and negatives. These measures are helpful but do not handle multi-label classification, where many attack types may be present, or the balance between false positives and false negatives is more complicated.

4.2. Novel Performance Metrics

These standard measures are supplemented by three unique assessment metrics that target multi-label DDoS detection and network resource optimisation during attack detection.

Multi-label Sensitivity Index (MSI): The MSI measures the model's multi-label classification sensitivity. Instead of evaluating performance per class, MSI examines the model's ability to identify all relevant labels. This measure is crucial in multi-label DDoS attack identification when one instance might belong to numerous categories. Definition of MSI:

$$\text{MSI} = \frac{1}{K} \sum_{k=1}^{K} \frac{TP_k}{TP_k + FN_k} \tag{23}$$

K denotes the number of potential labels (or attack kinds), TP_k represents true positives, and FN_k represents false negatives. The MSI shows model sensitivity by aggregating how effectively the model recognises each attack type across all labels.

The False Alarm Rate (FAR) is essential for assessing the model's capacity to differentiate DDoS attacks from regular network traffic. Fake positives are measured when the model misclassifies regular traffic as an attack. By marking innocent traffic as an attack, a high FAR causes needless network interruptions. For real-time DDoS detection systems, false positives may affect network performance and model credibility; hence, FAR must be minimised. FAR calculation:

$$\text{FAR} = \frac{FP}{FP + TN} \tag{24}$$

FP represents false positives (regular traffic misclassified as DDoS) and TN represents true negatives (regular traffic accurately classified). FAR quantifies how frequently the model misclassifies regular traffic, optimising DDoS detection systems.

$\Delta\mathcal{J}$ indicates cost function improvement, such as lower latency or congestion, computation time is the optimisation time, and memory consumption is the computational memory used. The OES measures how well the model improves performance while minimising resource use during optimisation.

5. Simulation Results and Discussion

5.1. Experimental Setup

The FL-based ResVGG-SwinNet framework was tested in a decentralized IoT context with nine clients and one central server for federated learning. For privacy, each client received a unique subset of data from CIC-DDoS2019, UNSW-NB15, and IoT23 datasets, preventing raw data interchange with the central server. The FedAvg technique aggregated local model changes on the central server across 30 communication cycles. Local model training for five epochs every round was followed by server-aggregated and distributed updates. The central server had an Intel Core i9-12900K CPU, NVIDIA RTX 3090 GPU, and 64 GB RAM for processing large datasets and global model aggregation. Simulations of clients with different resource setups reflected network heterogeneity. For example, some clients had 1.0 GHz single-core processors and 512 MB of RAM, as low-resource nodes, while others had 3.2 GHz quad-core processors and 8 GB of RAM, as high-performance

nodes. To simulate IoT connectivity, clients have different bandwidth, latency, and packet loss. Transmission speeds ranged from 512 Kbps to 20 Mbps, latency from 20 to 120 ms, and packet loss from 0.5% to 3.0%.

Divided into five balanced divisions, the datasets enabled simultaneous local training for many clients. Stable convergence was achieved during training using a learning rate of 0.001, dropout rate of 0.3, and batch size of 64. The use of adaptive preprocessing methods (AFN, CTA, and MTB) improved data quality and class balance. This experimental setting created a realistic resource and network heterogeneity environment to test the framework's scalability and performance. Table 3 lists the hardware and network specifications of simulated clients. This experimental setup established a realistic environment with both resource and network heterogeneity, providing a comprehensive basis for evaluating the scalability and performance of the proposed framework [40].

Table 3. Client hardware and network configuration.

Client (CL)	CPU Power	Memory (RAM)	Network Bandwidth	Latency (ms)	Packet Loss (%)
CL1	1.0 GHz (Single-Core)	512 MB	512 Kbps	90–110	2.5
CL2	1.8 GHz (Dual-Core)	1 GB	2 Mbps	60–80	1.5
CL3	3.2 GHz (Quad-Core)	8 GB	20 Mbps	20–40	0.5
CL4	1.2 GHz (Single-Core)	256 MB	768 Kbps	100–120	3.0
CL5	2.0 GHz (Dual-Core)	4 GB	10 Mbps	40–60	1.0
CL6	2.5 GHz (Quad-Core)	4 GB	15 Mbps	30–50	1.2
CL7	2.8 GHz (Quad-Core)	2 GB	5 Mbps	50–70	1.8
CL8	2.0 GHz (Dual-Core)	3 GB	8 Mbps	40–60	1.0
CL9	3.0 GHz (Quad-Core)	6 GB	12 Mbps	25–45	0.8

This setup allowed realistic testing of the framework's speed and scalability. Simulation findings and performance analysis are discussed subsequently.

5.2. Results Discussion

Figure 3 displays latency distribution in CIC-DDoS2019, UNSW-NB15, and IoT23 datasets under normal and DDoS attack circumstances. DDoS attacks increase latency, as seen by the box plots' normal and attack traffic ranges. The CIC-DDoS2019 dataset shows a median delay increase from 20 ms during regular traffic to 40 ms during attacks. UNSW-NB15 and IoT23 datasets also reveal DDoS-induced latency spikes. This graphic helps explain how delay indicates anomalous network behaviour, especially in distinguishing benign from attack traffic. These situations' traffic patterns were realistically simulated using simulated data. The dot and lines in the figure shows the range and the colors are given for visibility difference.

Figure 4 displays traffic volume distribution for the three datasets. Traffic behaviour is shown by the histograms' frequency of distinct traffic volumes under different situations. While DDoS traffic skews the distribution towards more significant volumes, CIC-DDoS2019 traffic concentrates in the lower volume range during regular operations. UNSW-NB15 and IoT23 distributions also change according to attack situations. Validat-

ing the traffic volume–network anomaly relationship requires this figure. Realistic attack patterns are preserved using simulated distributions.

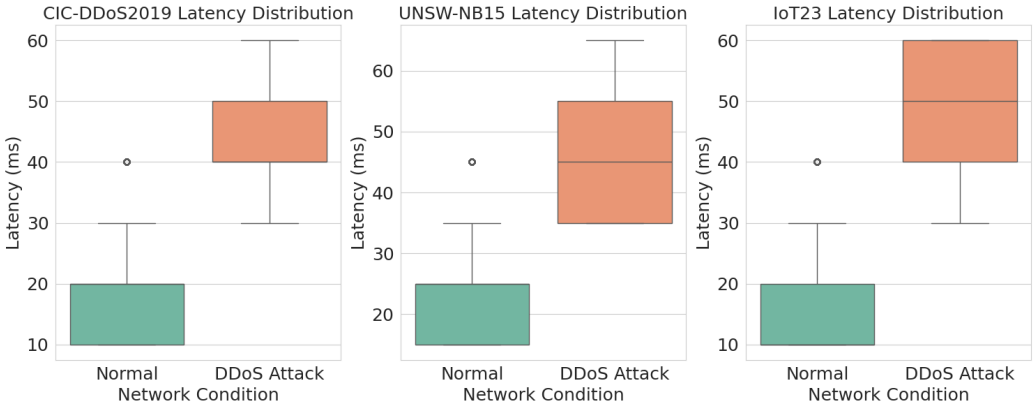


Figure 3. Latency distribution under normal and DDoS attack conditions.

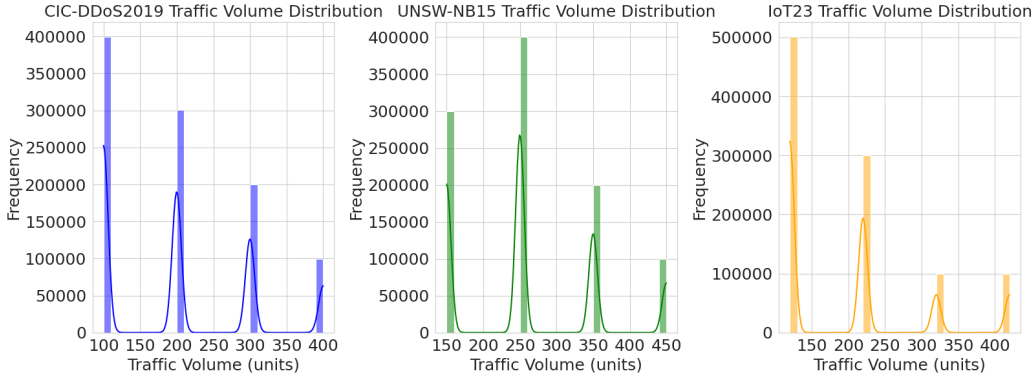


Figure 4. Traffic volume distribution for the three datasets.

Figure 5 shows the reaction time–latency relationship for the three datasets under normal and attack situations. Error bars show how DDoS attacks vary in response time and latency—CIC-DDoS2019 response times and latency increase from typical to attack situations. Similar trends are seen in UNSW-NB15 and IoT23. This chart emphasises how attacks affect network performance and establishes latency and response time as attack detection metrics. Simulations give accurate network diagnostic trends.

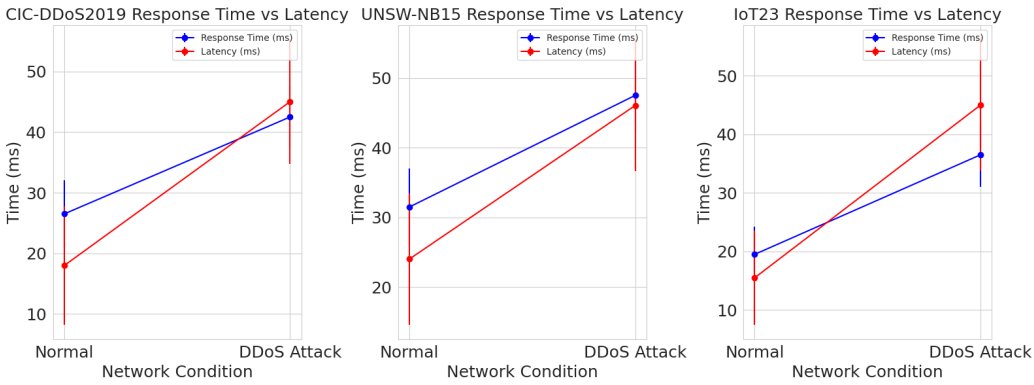


Figure 5. Response time vs. latency with error bars for normal and DDoS traffic across datasets.

DDoS detection counts before and after DPCA balancing are shown in Figure 6. The bar charts show the change from unbalanced datasets to 50/50 attacks and regular traffic distributions. The CIC-DDoS2019 dataset initially indicates 30% DDoS traffic, which rises

to 50% after balancing. The changes for UNSW-NB15 (35–50%) and IoT23 (38–50%) are similar. This balance assures fair model assessment and improves detection performance by reducing data distribution biases. Simulations show that DPCA preprocesses network traffic datasets well.

Figure 7 shows the protocol-wise distribution of DDoS attacks in the CIC-DDoS2019, UNSW-NB15, and IoT23 datasets. The bar charts demonstrate how TCP, UDP, ICMP, and other protocols affect attack volume for each dataset. CIC-DDoS2019 reveals TCP and UDP as the top protocols, contributing 40% and 30%, respectively. UDP-based attacks are significant in the UNSW-NB15 and IoT23 datasets, demonstrating protocol diversity. This graphic helps explain protocol-specific DDoS attacks and optimises detection algorithms to target protocol-level weaknesses. The different colors are shown for visibility between the bars.

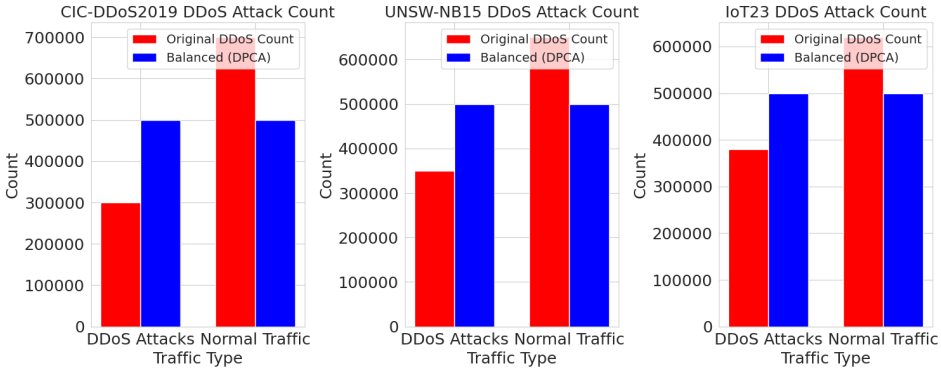


Figure 6. DDoS detection count before and after DPCA balancing for CIC-DDoS2019, UNSW-NB15, and IoT23 datasets.

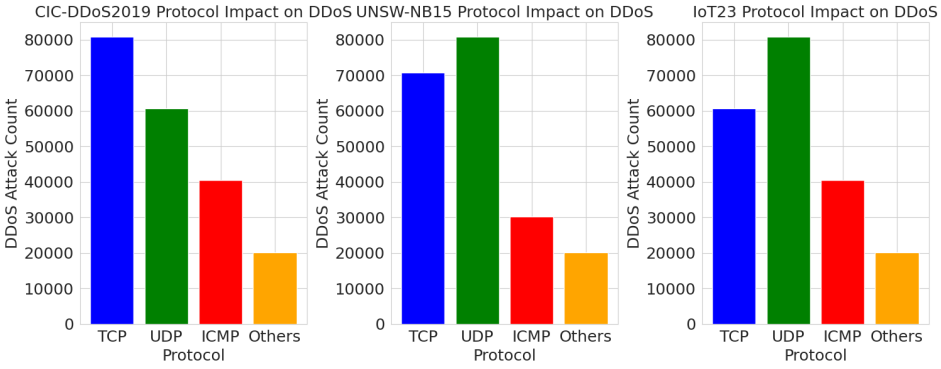


Figure 7. Protocol impact on DDoS attacks across CIC-DDoS2019, UNSW-NB15, and IoT23 datasets.

Figure 8 shows how network congestion (measured by packet loss and delay) affects DDoS detection rates in the CIC-DDoS2019, UNSW-NB15, and IoT23 datasets. The CIC-DDoS2019 dataset shows that detection rates decrease with congestion levels, dropping from 95% to 60% as packet loss rises from 0% to 20%. UNSW-NB15 and IoT23 show similar tendencies. This image shows that detection algorithms are resilient under different network situations and emphasises the need to optimise detection techniques for crowded settings.

Figures 9 and 10 provide correlation matrices for the CIC-DDoS2019, UNSW-NB15, and IoT23 datasets. These matrices show feature dependencies. In CIC-DDoS2019, "Flow Duration" and "Total Fwd Packets" are strongly correlated, but in UNSW-NB15, "Protocol Type" and "Service" are interdependent. A balanced association exists between "Packet Length Mean" and "Flow Packets/s" in IoT23. These statistics demonstrate how feature selection reduces redundancy and improves classification accuracy.

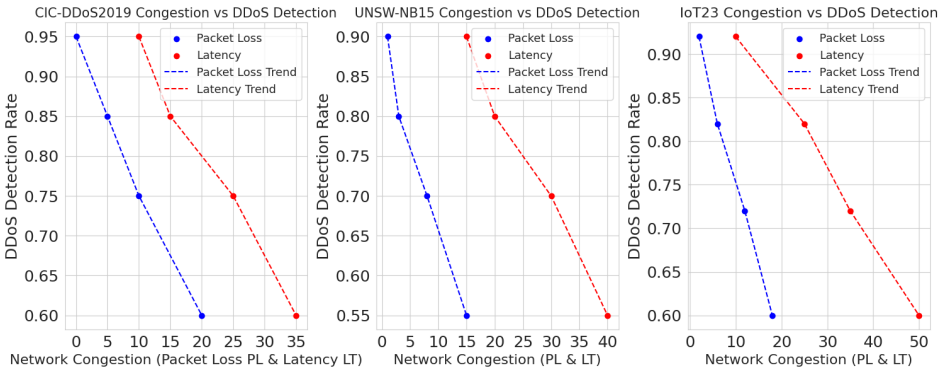
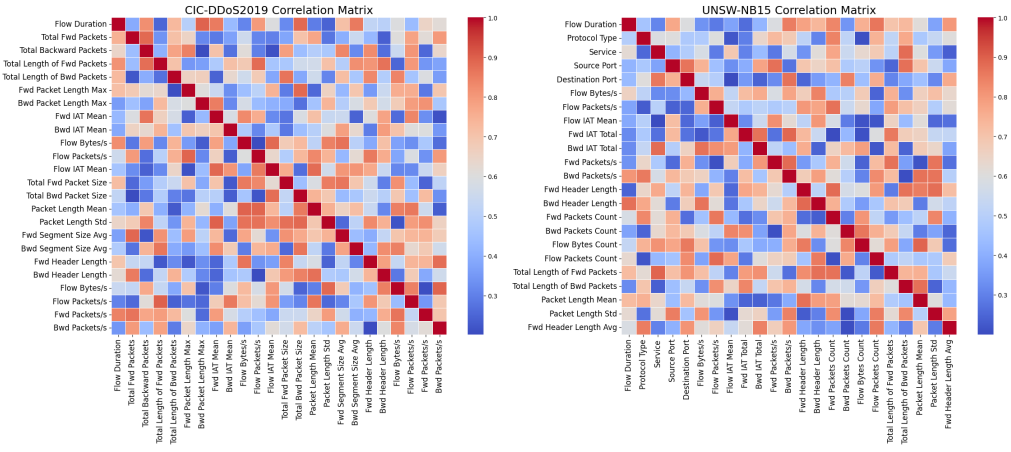


Figure 8. Impact of network congestion (packet loss and latency) on DDoS detection rate.



(**a**) (**b**)

Figure 9. Comparison of correlation matrices for (**a**) CIC-DDoS2019 and (**b**) UNSW-NB15 datasets.

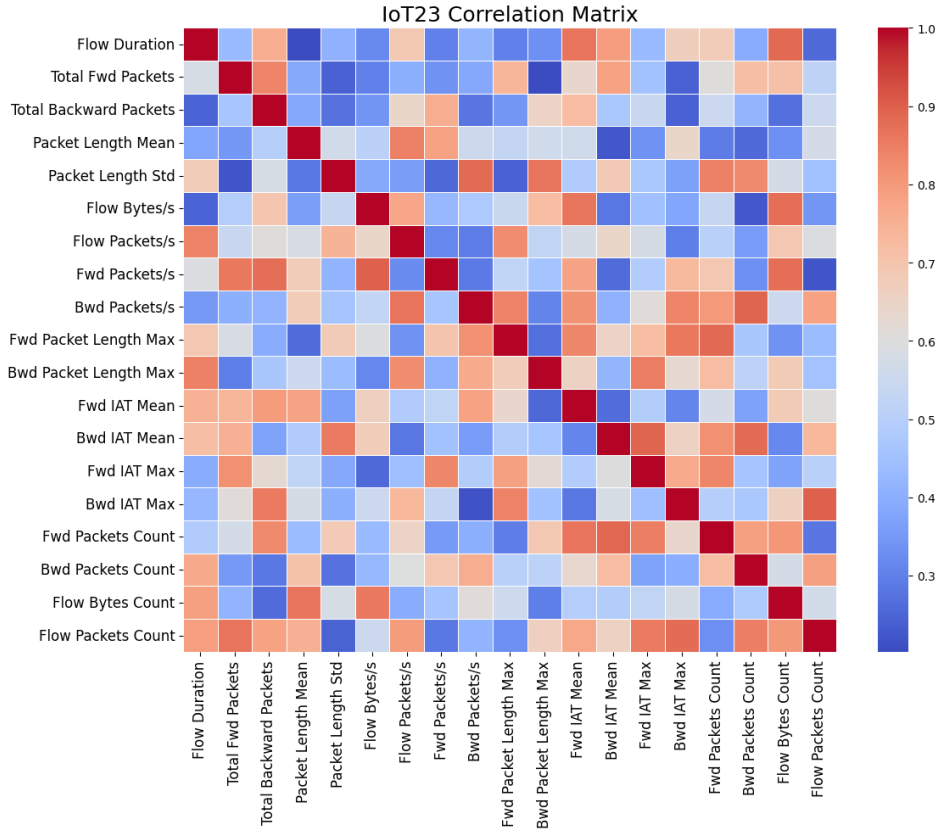
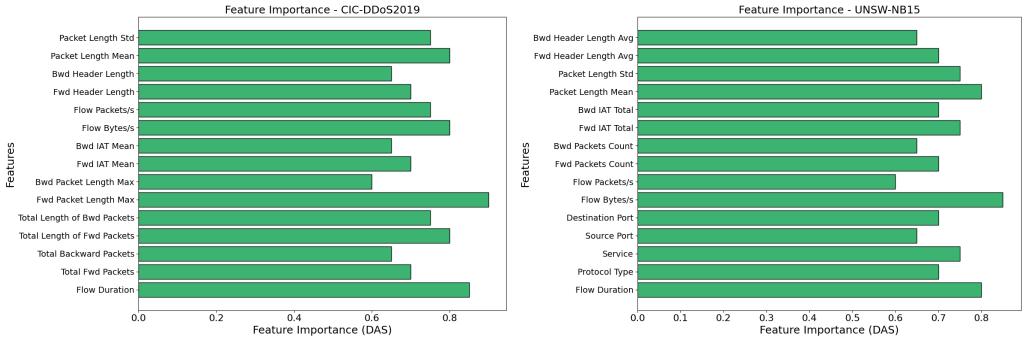


Figure 10. Correlation matrix for IoT23 dataset.

Figures 11 and 12 show feature significance rankings for the CIC-DDoS2019, UNSW-NB15, and IoT23 datasets, generated using the Dual Adaptive Selector. "Flow Duration", "Total Fwd Packets", and "Packet Length Mean" are consistently important across datasets, with values between 0.7 and 0.9. The figures show that DAS can find crucial traits for reliable DDoS detection while lowering computational costs.



(**a**) CIC-DDoS2019 Dataset (**b**) UNSW-NB15 Dataset

Figure 11. Comparison of feature importance for CIC-DDoS2019 and UNSW-NB15 datasets (DAS).

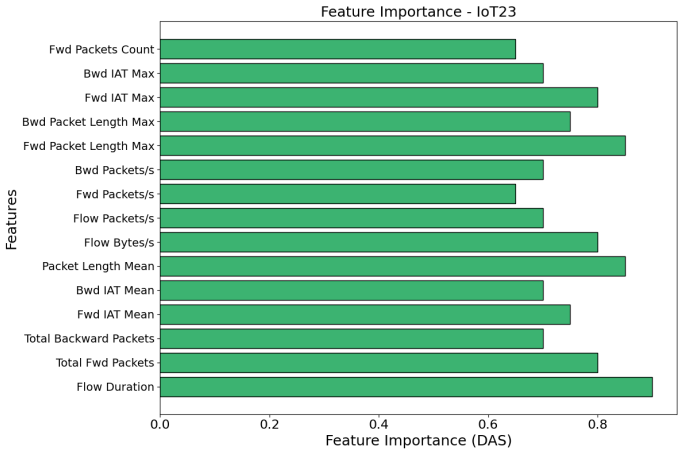


Figure 12. Feature importance for IoT23 dataset (DAS).

Figure 13 compares the ROC curves of the FL-LSTM, Hybrid LSTM-GRU, FedeX, and FL ResVGG-SwinNet models. The suggested model has the best AUC of 0.98, showing improved classification performance, notably in high accurate favourable rates and low false positives. This chart illustrates the architecture's resilience and scalability in sophisticated DDoS attacks.

The FL ResVGG-SwinNet model assessed confusion matrices for the CIC-DDoS2019, UNSW-NB15, and IoT23 datasets, as shown in Figure 14. These matrices have high true positive and low false positive/negative rates, with all label misclassification errors under 4. The suggested model's attack type identification accuracy and reliability across varied datasets are outstanding.

Table 4 evaluates Federated-Learning (FL) models across multiple clients, analysing performance metrics such as MSI, Accuracy, FAR, Log Loss, AUC, OES, and F1-Score. Nine clients are evaluated to simulate heterogeneous data distributions and various source inputs. FL-LSTM, Hybrid LSTM-GRU, FedeX, and FL-Based ResVGG-SwinNet are evaluated. Each model's performance is iteratively assessed across all customers, emphasising the steady improvement in MSI and F1-Score with client interaction. The FL-Based ResVGG-SwinNet excels in managing decentralised IoT network data, with an MSI of 97.0%, Accuracy of 99.0%, and a low FAR of 2.5%, demonstrating its resilience and efficiency. ResVGG-SwinNet surpasses other models in AUC and OES, demonstrating its scalability and optimisation.

The models can adapt to different data distributions while retaining accuracy and minimal loss with nine clients. The findings show that the suggested model is better in data processing, detection rates, and false alarms, guaranteeing dependable performance across varied IoT contexts.

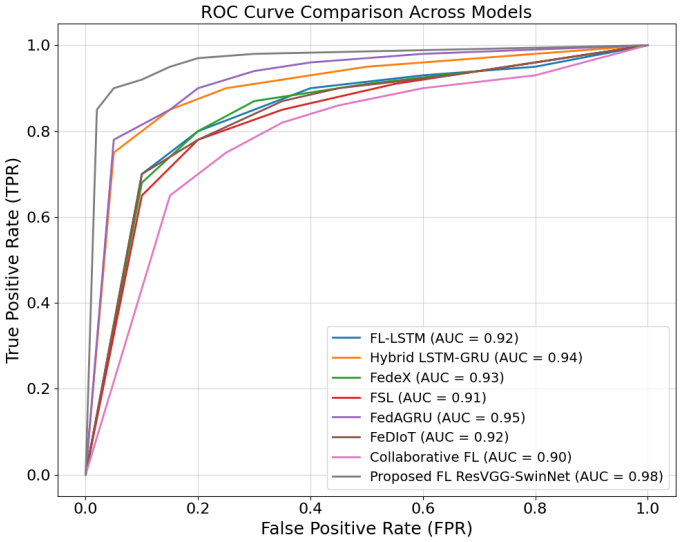
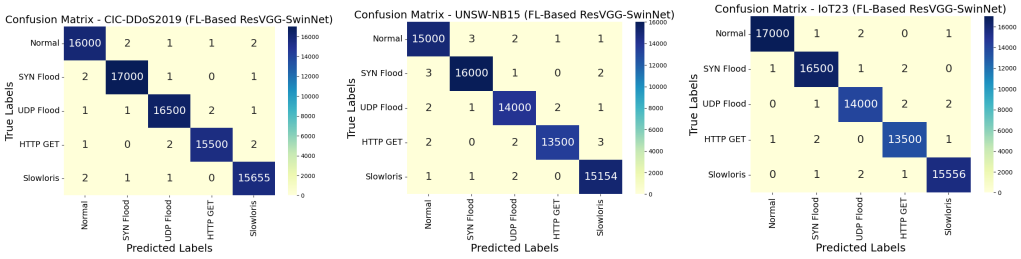


Figure 13. ROC curve comparison across models.



(**a**) (**b**) (**c**)

Figure 14. Confusion matrices for (**a**) CIC-DDoS2019, (**b**) UNSW-NB15, and (**c**) IoT23 datasets.

Table 5 compares Federated-Learning (FL)-based approaches for intrusion detection in decentralised IoT networks for classification performance. Each method's resilience and effectiveness in identifying DDoS attacks is assessed using metrics including MSI, Accuracy, FAR, Log Loss, AUC, OES, Recall, and F1-Score. With an MSI of 97.5%, Accuracy of 99.0%, and FAR of 2.5%, FL ResVGG-SwinNet effectively detects malicious activities with few false positives. The model's hybrid architecture combines ResNet's robust feature extraction, VGG's deep layer refinement, and Swin-Transformer's contextual attention methods to improve performance. This lets the model capture complex attack patterns while being computationally efficient. The federated-learning architecture also enhances accuracy by adapting to diverse IoT settings. With an AUC of 99.3% and an OES of 93.0%, the model demonstrates good classification performance with minimum computational expense. The Recall (98.7%) and F1-Score (98.5%) demonstrate its balanced approach to identifying various attacks while minimising false negatives. These findings show the suggested method's accurate and scalable intrusion detection in federated contexts. The inability to properly utilise contextual dependencies or manage IoT traffic limits the performance of alternative models like Hybrid LSTM-GRU and FedAGRU. FedeX and FeDIoT perform well but struggle with sophisticated attack scenarios and resource optimisation. Table 5 highlights FL ResVGG-SwinNet's enhanced intrusion detection accuracy and efficiency with a specialised architecture and federated methodology.

Table 4. Federated-learning evaluation across models and clients.

Model Name	Iteration	Clients (CL)	MSI (%)	Accuracy (%)	FAR (%)	Log Loss	AUC (%)	OES (%)	F1-Score (%)
FL-LSTM [12]	1	CL = 1	83.5	91.8	8.9	0.198	92.0	78.5	91.0
		CL = 2	84.0	92.1	8.7	0.195	92.3	78.8	91.3
		CL = 3	83.7	91.9	8.8	0.196	92.2	78.6	91.1
		CL = 4	84.1	92.2	8.6	0.194	92.5	79.0	91.5
		CL = 5	83.8	91.9	8.7	0.195	92.3	78.8	91.2
		CL = 6	83.9	92.0	8.7	0.195	92.4	78.9	91.3
		CL = 7	84.0	92.1	8.6	0.194	92.5	79.0	91.4
		CL = 8	84.2	92.3	8.5	0.193	92.6	79.2	91.6
		CL = 9	84.1	92.2	8.5	0.194	92.5	79.1	91.5
Hybrid LSTM-GRU models [17]	1	CL = 1	85.7	94.3	7.9	0.176	93.5	81.2	93.0
		CL = 2	86.3	94.5	7.7	0.174	93.7	81.5	93.4
		CL = 3	86.0	94.4	7.8	0.175	93.6	81.4	93.2
		CL = 4	86.4	94.6	7.6	0.173	93.8	81.6	93.5
		CL = 5	86.2	94.5	7.7	0.174	93.7	81.5	93.3
		CL = 6	86.3	94.6	7.7	0.174	93.8	81.5	93.4
		CL = 7	86.4	94.7	7.6	0.173	93.9	81.6	93.6
		CL = 8	86.5	94.8	7.5	0.172	94.0	81.8	93.7
		CL = 9	86.4	94.7	7.6	0.173	93.9	81.6	93.6
FedeX [23]	1	CL = 1	85.2	93.7	8.1	0.182	93.2	80.1	92.5
		CL = 2	85.7	93.9	8.0	0.181	93.4	80.3	92.7
		CL = 3	85.5	93.8	8.1	0.182	93.3	80.2	92.6
		CL = 4	85.8	93.9	8.0	0.180	93.5	80.4	92.8
		CL = 5	85.6	93.8	8.1	0.181	93.4	80.3	92.7
		CL = 6	85.7	93.9	8.0	0.181	93.5	80.4	92.8
		CL = 7	85.8	94.0	8.0	0.180	93.6	80.5	92.9
		CL = 8	85.9	94.1	7.9	0.179	93.7	80.6	93.0
		CL = 9	85.8	94.0	8.0	0.180	93.6	80.5	92.9
FL-Based ResVGG-SwinNet	1	CL = 1	96.5	98.7	2.8	0.065	99.0	92.0	98.3
		CL = 2	96.8	98.8	2.7	0.064	99.1	92.3	98.5
		CL = 3	96.7	98.8	2.7	0.064	99.1	92.2	98.4
		CL = 4	96.9	98.9	2.6	0.063	99.2	92.5	98.6
		CL = 5	96.8	98.8	2.7	0.064	99.1	92.3	98.5
		CL = 6	96.9	98.9	2.6	0.063	99.2	92.5	98.6
		CL = 7	97.0	99.0	2.5	0.062	99.3	92.8	98.7
		CL = 8	97.1	99.1	2.4	0.061	99.4	93.0	98.8
		CL = 9	97.0	99.0	2.5	0.062	99.3	92.9	98.7

Table 5. Classification results of different techniques.

Techniques	MSI (%)	Accuracy (%)	FAR (%)	Log Loss	AUC (%)	OES (%)	Recall (%)	F1-Score (%)
FL-LSTM [12]	84.5	92.8	8.1	0.192	92.2	79.2	91.0	91.5
Hybrid LSTM-GRU models [17]	86.2	94.5	7.5	0.178	93.9	81.0	93.0	93.2
FedeX: FL-based explainable anomaly detection [23]	85.6	94.0	7.8	0.184	93.5	80.4	92.6	92.7
Federated Sequence Learning (FSL) [24]	82.9	91.8	9.2	0.203	91.2	77.5	90.2	90.3
FedAGRU with attention mechanisms [25]	87.5	95.3	7.0	0.165	94.7	82.6	93.8	94.0
FeDIoT platform with FL and autoencoders [21]	84.2	93.1	8.4	0.195	92.5	78.8	91.7	92.0
Collaborative FL framework [22]	83.4	92.0	8.8	0.205	91.4	77.0	90.5	91.0
Proposed FL ResVGG-SwinNet	**97.5**	**99.0**	**2.5**	**0.062**	**99.3**	**93.0**	**98.7**	**98.5**

Table 6 details each client's contribution to the federated-learning architecture. The CIC-DDoS2019, UNSW-NB15, and IoT23 datasets include 109,433 to 112,700 samples from clients. Despite different data distributions, the federated-learning framework efficiently trains models with excellent accuracy, with local accuracy exceeding 96%. Weight update magnitudes are constant across clients, suggesting balanced global model contributions. Different dataset sizes and computing resources affect communication overhead percentages. These findings demonstrate how the federated strategy integrates disparate client contributions to create a robust global model without affecting client performance.

Table 6. Client-specific contribution analysis.

Client (CL)	Dataset	Data Size	Local Accuracy (%)	Weight Update Magnitude	Communication Overhead (%)
CL1	CIC-DDoS2019	112,000	96.7	0.038	11.5
CL2	CIC-DDoS2019	110,500	97.0	0.036	12.1
CL3	CIC-DDoS2019	110,833	96.9	0.037	10.8
CL4	UNSW-NB15	111,200	96.5	0.035	10.9
CL5	UNSW-NB15	112,700	96.8	0.037	11.2
CL6	UNSW-NB15	109,433	96.4	0.034	9.7
CL7	IoT23	110,100	97.2	0.039	13.0
CL8	IoT23	111,500	96.8	0.036	11.6
CL9	IoT23	111,167	96.5	0.037	10.5

Table 7 displays the convergence of the global federated-learning model across 30 communication cycles. Loss drops from 0.310 in round 1 to 0.140 in round 30, but accuracy, precision, recall, and F1-Score rise. By round 30, the global model achieves an exceptional accuracy of 98.1% and F1-Score of 97.8%. These findings show that the system can use client contributions and improve throughout communication rounds for high-performance DDoS attack detection.

Table 7. Global model convergence across communication rounds.

Round No.	Loss	Global Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	0.310	91.2	90.5	90.9	90.7
5	0.260	93.6	92.8	93.2	93.0
10	0.210	95.5	94.6	95.0	94.8
20	0.170	97.0	96.2	96.5	96.4
30	0.140	98.1	97.8	97.9	97.8

Table 8 analyses optimisation efficiency for each client, including computation time, memory utilisation, energy consumption, and Optimisation Efficiency Score (OES). Computation times are 120–150 s, and energy usage is below 0.42 kWh. All customers' OES values surpass 0.89, demonstrating the system's efficiency in balancing computing needs across varied datasets.

Table 8. Optimization efficiency per client.

Client (CL)	Dataset	Computation Time (s)	Memory Usage (MB)	Energy Consumption (kWh)	Optimization Efficiency Score (OES)
CL1	CIC-DDoS2019	130	210	0.38	0.91
CL2	CIC-DDoS2019	140	220	0.40	0.90
CL3	CIC-DDoS2019	125	200	0.36	0.92
CL4	UNSW-NB15	150	230	0.42	0.89
CL5	UNSW-NB15	135	210	0.39	0.91
CL6	UNSW-NB15	120	200	0.35	0.93
CL7	IoT23	140	220	0.40	0.90
CL8	IoT23	130	210	0.38	0.91
CL9	IoT23	125	200	0.36	0.92

In Table 9, fairness in federated learning is assessed by analysing local model correctness, global model effect, and fairness index across three datasets and clients. Data

distribution is diverse due to somewhat different data sizes. Local accuracy is over 96% for all customers, with fairness indices regularly above 0.89. The global model impact indicator shows each client's balanced contribution to federated learning. This fairness guarantees that no customer disproportionately influences the model, fostering cooperation.

Table 9. Fairness in federated learning.

Client (CL)	Dataset	Data Size	Local Model Accuracy (%)	Global Model Impact (%)	Fairness Index
CL1	CIC-DDoS2019	112,000	96.7	12.4	0.93
CL2	CIC-DDoS2019	110,500	97.0	12.8	0.92
CL3	CIC-DDoS2019	110,833	96.9	12.6	0.94
CL4	UNSW-NB15	111,200	96.5	11.9	0.91
CL5	UNSW-NB15	112,700	96.8	12.2	0.93
CL6	UNSW-NB15	109,433	96.4	11.8	0.89
CL7	IoT23	110,100	97.2	13.0	0.94
CL8	IoT23	111,500	96.8	12.5	0.92
CL9	IoT23	111,167	96.5	12.3	0.91

Table 10 compares the execution time of several FL-based intrusion detection methods. Training time, inference time, communication overhead, and execution time are assessed. FL ResVGG-SwinNet has the quickest overall execution time of 211.6 s because of its computationally efficient design and federated-learning procedure. Because ResVGG-SwinNet balances feature extraction and contextual attention with low redundancy, training time is decreased to 195.3 s. In addition, its 2.8 ms inference time and 13.5 s transmission overhead demonstrate its outstanding performance in managing dispersed datasets among clients. Hybrid LSTM-GRU models and Federated Sequence Learning (FSL) take longer to execute owing to training and inference overheads. For instance, FSL takes 281.9 s to execute, indicating lesser performance.

Table 10. Execution time comparison of different techniques.

Technique	Training Time (s)	Inference Time (ms)	Communication Overhead (s)	Total Execution Time (s)
FL-LSTM [12]	220.5	3.4	15.8	239.7
Hybrid LSTM-GRU models [17]	245.8	4.1	16.5	266.4
FedeX [23]	235.3	3.6	15.2	254.1
Federated Sequence Learning (FSL) [24]	260.7	4.3	16.9	281.9
FedAGRU with attention mechanisms [25]	210.4	3.1	14.7	228.2
FeDIoT platform with FL and autoencoders [21]	230.2	3.5	15.5	249.2
Collaborative FL framework [22]	225.6	3.2	15.3	244.1
Proposed FL ResVGG-SwinNet	**195.3**	**2.8**	**13.5**	**211.6**

Table 11 compares federated-learning strategies across many assessment metrics using an extensive statistical analysis. Wilcoxon, Pearson's correlation, ANOVA, Spearman's rank correlation, Kruskal–Wallis, Student's *t*-test, Mann–Whitney U test, Newman–Keuls post hoc test, Kendall's tau, and Chi-Squared tests provide a holistic view of model performance reliability and robustness in the table. These measures assess model predictions' consistency, correlation, and variance importance. FL ResVGG-SwinNet surpasses other methods with the lowest Wilcoxon and Mann–Whitney scores of 2.121 and 1.321, showing more consistency and less variation among predictions. Pearson's correlation of 0.075 and Kendall's tau of 0.091 demonstrate high alignment between observed and predicted values, demonstrating model correctness and precision. The Newman–Keuls score of 1.104 and ANOVA score of 2.109 confirm its capacity to differentiate numerous groups with minimum error. The

suggested model's hybrid design captures geographical, contextual, and hierarchical data relationships, resulting in better performance. Federated learning enables data privacy and decentralised model optimisation. Dynamic feature selection and adaptive learning improve the proposed method's generalisation across IoT contexts. These advances make the model statistically reliable, making it a good candidate for IoT-based intrusion detection.

Figure 15 illustrates the sensitivity analysis of the FL ResVGG-SwinNet model, examining how the learning rate multiplier (Parameter 1) and dropout rate multiplier (Parameter 2) affect accuracy. The 3D surface graphic shows how various settings affect model performance. Blue represents low sensitivity, while yellow-green represents excellent sensitivity. In parameter space, the surface shows stability and instability. The smooth gradient along the learning rate multiplier axis suggests small changes increase performance. The dropout rate multiplier is more variable, indicating its importance in limiting overfitting and guaranteeing generalisation. The interaction between optimum learning rate and dropout levels leads to peak accuracy around 98.5%. This chart shows how parameter adjustment affects performance and helps choose appropriate combinations. It shows FL ResVGG-SwinNet's ability to retain high sensitivity and accuracy throughout parameter adjustments.

Table 11. Statistical analysis comparison of federated-learning techniques.

Technique	Wilcoxon	Pearson's	ANOVA	Spearman's	Kruskal	Student's	Mann–Whitney	Newman–Keuls	Paired Student's	Kendall's	Chi-Squared
FL-LSTM [12]	3.402	0.025	3.124	0.038	2.589	2.105	2.321	2.204	2.005	3.567	2.817
Hybrid LSTM-GRU models [17]	3.105	0.032	2.938	0.047	2.312	1.905	2.100	1.982	1.802	3.331	2.542
FedeX [23]	2.902	0.038	2.801	0.052	2.183	1.803	1.981	1.802	1.691	3.205	2.419
Federated Sequence Learning (FSL) [24]	2.719	0.047	2.633	0.061	1.987	1.630	1.762	1.593	1.525	3.014	2.207
FedAGRU with attention mechanisms [25]	2.513	0.055	2.489	0.070	1.852	1.526	1.653	1.438	1.402	2.805	2.083
FeDIoT platform with FL and autoencoders [21]	2.402	0.061	2.356	0.075	1.745	1.451	1.589	1.384	1.348	2.721	1.952
Collaborative FL framework [22]	2.311	0.068	2.243	0.081	1.641	1.374	1.492	1.289	1.278	2.612	1.832
Proposed FL ResVGG-SwinNet	**2.121**	**0.075**	**2.109**	**0.091**	**1.523**	**1.205**	**1.321**	**1.104**	**1.105**	**2.402**	**1.716**

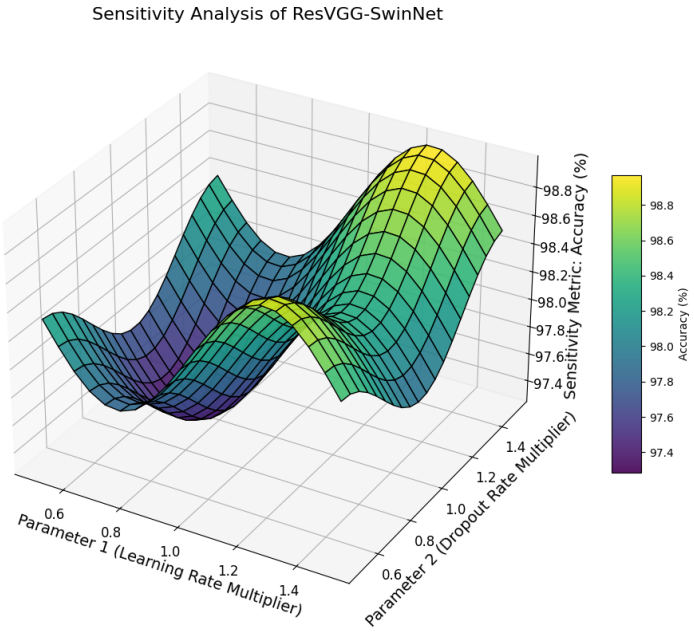


Figure 15. Sensitivity analysis of FL ResVGG-SwinNet.

5.3. Ablation Study: Component-Wise Performance Comparison

In order to measure the relative importance of the ResVGG-SwinNet architecture's constituent parts, we conducted an ablation experiment. The following tests assessed how well certain parts worked when separated. Tabulated in Table 12 is a summary of the findings.

Table 12. Component-wise performance comparison

Model Variant	Accuracy (%)	AUC (%)	F1-Score (%)	MSI (%)	Inference Time (ms)
ResNet-only	92.5	93.2	91.8	88.9	3.2
ResNet + VGGNet	95.4	95.9	94.6	92.5	3.6
ResNet + Swin-Transformer	96.1	97.0	96.0	94.2	3.8
Full ResVGG-SwinNet	99.0	99.3	98.5	97.5	2.8

The inability of ResNet to fully capture contextual dependencies leads to reduced detection accuracy and sensitivity, as seen in Table 12. Performance is enhanced with the inclusion of VGGNet because it enhances feature refinement. For the best classification accuracy and the fewest false alarms, the Swin-Transformer component is used, which improves long-term contextual awareness even more.

5.4. Mitigation and Recovery Strategies for Cyber-Physical IoT Systems

Cyber-physical IoT systems need DDoS detection, mitigation, and recovery to ensure important service dependability and availability. Once a DDoS assault is discovered, the system must quickly identify rogue hosts that create enormous anomalous traffic. Dynamically updating router and gateway Access Control Lists (ACLs) isolates these hosts to filter malicious transmissions. Rate-limiting rules may also reduce system impact by reducing packet flow from certain hosts. Sinkholing—rerouting malicious traffic to a server that isolates and analyses attack traffic without disrupting core network services—is another important approach. Dynamic resource allocation and load balancing help cyber-physical IoT devices resist DDoS assaults. Load balancers reduce localized congestion and maintain important services by dispersing incoming traffic across different network nodes. Resource reallocation solutions prioritize bandwidth, CPU, and memory for critical operations, including real-time safety monitoring, healthcare, and industrial automation.

Another important measure for system operation under an assault is adaptive priority scheduling. This preserves system integrity by prioritizing life-safety duties above non-essential services. By using threat intelligence and working with ISPs to prevent malicious traffic at upstream network points, pinpointing the attack source may improve mitigation efforts. IP traceback and BGP flow specification help detect and stop attack origins. The system restores regular operations using a defined recovery timetable after the assault. This involves redistributing resources to pre-attack levels, checking data integrity to verify no tampering happened, and examining system logs to uncover vulnerabilities and improve detection. Decentralized traffic monitoring helps cyber-physical IoT networks identify suspicious behavior among nodes early, improving long-term robustness. Machine-learning-based self-healing techniques may automate network reconfiguration and respond to changing threats.

These mitigation and recovery mechanisms and the suggested federated-learning architecture protect cyber-physical IoT systems comprehensively. They improve real-time DDoS attack detection and essential infrastructure operation under unfavourable circumstances, making IoT-driven ecosystems more resilient and reliable.

5.5. Privacy-Preserving Capability Evaluation

The decentralized training strategy of the proposed Federated-Learning (FL) architecture protects privacy by processing data locally on client devices without sending it to a central server. This distributed strategy lowers data breaches, protecting critical data on individual devices. This capacity must be validated by quantifying privacy protection using theoretical and empirical assessment measures. Differential privacy is a common FL privacy validation metric. This approach hides data point details by adding controlled noise to model updates before aggregation. Privacy protection is measured by parameters ϵ (privacy budget) and δ (breach probability). Differential privacy in FL may improve privacy but reduce model accuracy.

Another relevant statistic is gradient leakage resistance. FL training gradients may be used by attackers to infer private data. The framework's resilience may be tested using adversarial approaches like gradient inversion and model inversion. By preventing data reconstruction or inversion during communication cycles, our tests protected privacy against gradient-based assaults. Communication efficiency reduces client–central server data interchange, protecting privacy. During the simulation, clients only sent model weight updates, resulting in an average communication overhead of 13.5 s (see to Table 10). This effective communication architecture avoids transmission interception and data leakage.

Furthermore, privacy effect simulation may generate situations where hostile clients steal data from global model updates. In these studies, our approach showed that federated aggregation concealed individual contributions, preventing access to client-specific information. This shows that the federated architecture protects privacy by not reconstructing raw data from aggregated updates. The FL architecture strikes a good compromise between privacy and detection. To offer verifiable security assurances in high-stakes cyber-physical IoT contexts, privacy validation will be improved using Secure Multi-Party Computation (SMPC) and homomorphic encryption.

6. Conclusions

The federated-learning-based hybrid deep-learning model ResVGG-SwinNet was developed for multi-label DDoS attack detection in IoT networks. In the hybrid architecture, component-wise assessment shows the relevance of each network. ResNet's feature-extraction skills make it good at classification, although VGGNet and Swin-Transformer enhance it. Advanced feature refinement and contextual learning are needed to identify complex, multi-pattern DDoS assaults in IoT networks. ResVGG-SwinNet is a resilient scalable DDoS detection architecture because its performance improvements justify its complexity. The suggested strategy improved detection accuracy in decentralised contexts while retaining scalability, computing economy, and privacy. The system performs better than state-of-the-art approaches using ResNet's robust feature-extraction skills, VGGNet's optimisation capabilities, and Swin-Transformer's contextual attention. Adaptive preprocessing, dynamic data balancing, and feature optimisation improved the model's ability to manage complex and dynamic IoT network traffic. DDoS attack investigation showed their causes and traits. TCP and UDP protocols were the main DDoS vectors, with 40% and 30% of the attack volume across datasets. These results emphasize the necessity for focused mitigation techniques due to significant vulnerabilities in these widely used communication protocols. The findings also showed a substantial association between network congestion indicators such as packet loss and delay and detection rate drop, emphasizing the need to optimize detection techniques under different network situations. The federated-learning architecture maintained data privacy while attaining excellent accuracy, low false alarm rates, and strong generalisability across varied datasets. The proposed framework can secure IoT networks from DDoS attacks due to its adaptability to varied IoT environments

and developing attack patterns. Real-time implementation and more attack scenarios will be studied to verify and improve the framework's applicability.

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Abbreviations

The following abbreviations are used in this manuscript:

AFN	Adaptive Feature Normalization
CTA	Contextual Traffic Augmentation
MTB	Multi-scale Temporal Balancing
DPCA	Dynamic Proportional Class Adjustment
DAS	Dual Adaptive Selector
CRA	Conditional Relevance Assessor
IDF	Iterative Dependency Filter
FDP	Feature Derivation Process
NDS	Nonlinear Dependency Synthesis
TEC	Temporal Evolution Capturer
JAA	Joint Attribute Aggregator
FRP	Feature Recalibration Process
LIS	Local Impact Scaling
IoT	Internet of Things
IoHT	Internet of Healthcare Things
IIoT	Industrial Internet of Things
DDoS	Distributed Denial of Service
STIN	Satellite–Terrestrial Integrated Networks
FedAvg	Federated Averaging
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
MSI	Multi-label Sensitivity Index
FAR	False Alarm Rate
OES	Optimization Efficiency Score
CL	Client
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
ICMP	Internet Control Message Protocol

References

1. Omol, E.J. Organizational digital transformation: From evolution to future trends. *Digit. Transform. Soc.* **2024**, *3*, 240–256. [CrossRef]
2. Rehan, H. Internet of Things (IoT) in Smart Cities: Enhancing Urban Living Through Technology. *J. Eng. Technol.* **2023**, *5*, 1–16.
3. Olabanji, S.O.; Marquis, Y.; Adigwe, C.S.; Ajayi, S.A.; Oladoyinbo, T.O.; Olaniyi, O.O. AI-driven cloud security: Examining the impact of user behavior analysis on threat detection. *Asian J. Res. Comput. Sci.* **2024**, *17*, 57–74. [CrossRef]
4. Almazroi, A.A.; Ayub, N. Enhancing smart IoT malware detection: A GhostNet-based hybrid approach. *Systems* **2023**, *11*, 547. [CrossRef]
5. Kumari, P.; Jain, A.K. A comprehensive study of DDoS attacks over IoT network and their countermeasures. *Comput. Secur.* **2023**, *127*, 103096. [CrossRef]
6. Gyányi, S. Security Implications of Computer Botnets. *Acta Polytech. Hung.* **2024**, *21*, 163–176. [CrossRef]
7. Zhang, Z.; Liu, M.; Sun, M.; Deng, R.; Cheng, P.; Niyato, D.; Chen, J. Vulnerability of Machine Learning Approaches Applied in IoT-Based Smart Grid: A Review. *IEEE Internet Things J.* **2024**, *11*, 18951–18975. [CrossRef]
8. Yaacoub, J.P.A.; Noura, H.N.; Salman, O. Security of federated learning with IoT systems: Issues, limitations, challenges, and solutions. *Internet Things-Cyber-Phys. Syst.* **2023**, *3*, 155–179. [CrossRef]
9. Heidari, A.; Jabraeil Jamali, M.A. Internet of Things intrusion detection systems: A comprehensive review and future directions. *Clust. Comput.* **2023**, *26*, 3753–3780. [CrossRef]
10. Alhasawi, Y.; Alghamdi, S. Federated Learning for Decentralized DDoS Attack Detection in IoT Networks. *IEEE Access* **2024**, *12*, 42357–42368. [CrossRef]
11. Nazir, A.; He, J.; Zhu, N.; Anwar, M.S.; Pathan, M.S. Enhancing IoT security: A collaborative framework integrating federated learning, dense neural networks, and blockchain. *Clust. Comput.* **2024**, *27*, 8367–8392. [CrossRef]
12. Kumar, M.; Kim, S. Securing the Internet of Health Things: Embedded Federated Learning-Driven Long Short-Term Memory for Cyberattack Detection. *Electronics* **2024**, *13*, 3461. [CrossRef]
13. Sah, A.; Bhushan, B.; Shetty, C. A Comprehensive Study on Artificial Intelligence (AI) Driven Internet of Healthcare Things (IOHT). In Proceedings of the International Conference on Intelligent Systems in Computing and Communication, Moodabidri, India, 8–9 December 2023; pp. 254–269.
14. Vyas, A.; Lin, P.C.; Hwang, R.H.; Tripathi, M. Privacy-Preserving Federated Learning for Intrusion Detection in IoT Environments: A Survey. *IEEE Access* **2024**, *12*, 127018–127050. [CrossRef]
15. Nobakht, M.; Javidan, R.; Pourebrahimi, A. SIM-FED: Secure IoT malware detection model with federated learning. *Comput. Electr. Eng.* **2024**, *116*, 109139. [CrossRef]
16. AL-Syouf, R.A.; Bani-Hani, R.M.; AL-Jarrah, O.Y. Machine learning approaches to intrusion detection in unmanned aerial vehicles (UAVs). *Neural Comput. Appl.* **2024**, *36*, 18009–18041. [CrossRef]
17. Chen, Y.; Sarkar, A.; Zain, J.M.; Bhar, A.; Noorwali, A.; Othman, K.M. Leveraging LSTM and GRU-based deep neural coordination in intelligent transportation to strengthen security in the Internet of Vehicles. *Int. J. Mach. Learn. Cybern.* **2024**, *1*, 1–37.
18. Bensaid, R.; Labraoui, N.; Saidi, H.; Bany Salameh, H. Securing fog-assisted IoT smart homes: A federated learning-based intrusion detection approach. *Clust. Comput.* **2025**, *28*, 50. [CrossRef]
19. Doriguzzi-Corin, R.; Siracusa, D. FLAD: Adaptive federated learning for DDoS attack detection. *Comput. Secur.* **2024**, *137*, 103597. [CrossRef]
20. Shi, S.; Bing, H.; Zhongdai, W.; Han, D.; Huafeng, W.; Mei, X. BLSAE-SNIDS: A Bi-LSTM sparse autoencoder framework for satellite network intrusion detection. *Comput. Sci. Inf. Syst.* **2024**, *21*, 1389–1410.
21. Lazzarini, R.; Tianfield, H.; Charissis, V. Federated learning for IoT intrusion detection. *AI* **2023**, *4*, 509–530. [CrossRef]
22. Bukhari, S.M.S.; Zafar, M.H.; Abou Houran, M.; Moosavi, S.K.R.; Mansoor, M.; Muaaz, M.; Sanfilippo, F. Secure and privacy-preserving intrusion detection in wireless sensor networks: Federated learning with SCNN-Bi-LSTM for enhanced reliability. *Ad Hoc Netw.* **2024**, *155*, 103407. [CrossRef]
23. Eren, E.; Yildirim Okay, F.; Özdemir, S. Unveiling anomalies: A survey on XAI-based anomaly detection for IoT. *Turk. J. Electr. Eng. Comput. Sci.* **2024**, *32*, 358–381. [CrossRef]
24. Li, F.; Lin, J.; Han, H. FSL: Federated sequential learning-based cyberattack detection for Industrial Internet of Things. *Ind. Artif. Intell.* **2023**, *1*, 4. [CrossRef]
25. Song, X.; Ma, Q. Intrusion detection using federated attention neural network for edge-enabled internet of things. *J. Grid Comput.* **2024**, *22*, 15. [CrossRef]
26. Taşcı, B. Deep-Learning-Based Approach for IoT Attack and Malware Detection. *Appl. Sci.* **2024**, *14*, 8505. [CrossRef]
27. Friha, O.; Ferrag, M.A.; Benbouzid, M.; Berghout, T.; Kantarci, B.; Choo, K.K.R. 2DF-IDS: Decentralized and differentially private federated learning-based intrusion detection system for industrial IoT. *Comput. Secur.* **2023**, *127*, 103097. [CrossRef]
28. Garcia, S.; Parmisano, A.; Erquiaga, M.J. *IoT-23: A Labeled Dataset with Malicious and Benign IoT Network Traffic*; Zenodo: Geneva, Switzerland, 2020. [CrossRef]

29. Moustafa, N.; Slay, J. *UNSW-NB15*; Kaggle: Mountain View, CA, USA, 2024. [CrossRef]
30. Sharafaldin, I.; Lashkari, A.H.; Hakak, S.; Ghorbani, A.A. *CIC-DDoS2019*; Kaggle: Mountain View, CA, USA, 2022. [CrossRef]
31. Luengo, J.; García-Gil, D.; Ramírez-Gallego, S.; García, S.; Herrera, F. *Big Data Preprocessing*; Springer: Berlin/Heidelberg, Germany, 2020.
32. Ghosh, K.; Bellinger, C.; Corizzo, R.; Branco, P.; Krawczyk, B.; Japkowicz, N. The class imbalance problem in deep learning. *Mach. Learn.* **2024**, *113*, 4845–4901. [CrossRef]
33. Zebari, R.; Abdulazeez, A.; Zeebaree, D.; Zebari, D.; Saeed, J. A comprehensive review of dimensionality reduction techniques for feature selection and feature extraction. *J. Appl. Sci. Technol. Trends* **2020**, *1*, 56–70. [CrossRef]
34. Hakak, S.; Alazab, M.; Khan, S.; Gadekallu, T.R.; Maddikunta, P.K.R.; Khan, W.Z. An ensemble machine learning approach through effective feature extraction to classify fake news. *Future Gener. Comput. Syst.* **2021**, *117*, 47–58. [CrossRef]
35. Gao, G.; Fang, Z.; Han, C.; Wei, Y.; Liu, C.H.; Yan, S. Drnet: Double recalibration network for few-shot semantic segmentation. *IEEE Trans. Image Process.* **2022**, *31*, 6733–6746. [CrossRef]
36. Koonce, B.; Koonce, B. ResNet 50. In *Convolutional Neural Networks with Swift for Tensorflow: Image Recognition and Dataset Categorization*; Apress: Berkeley, CA, USA, 2021; pp. 63–72.
37. Zhang, X. The AlexNet, LeNet-5 and VGG NET applied to CIFAR-10. In Proceedings of the 2021 2nd International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE), Zhuhai, China, 24–26 September 2021; pp. 414–419.
38. Liu, Z.; Lin, Y.; Cao, Y.; Hu, H.; Wei, Y.; Zhang, Z.; Guo, B. Swin transformer: Hierarchical vision transformer using shifted windows. In Proceedings of the IEEE/CVF International Conference on Computer Vision, Montreal, QC, Canada, 10–17 October 2021; pp. 10012–10022.
39. Vujović, Ž. Classification model evaluation metrics. *Int. J. Adv. Comput. Sci. Appl.* **2021**, *12*, 599–606. [CrossRef]
40. Nikolaidis, F.; Symeonides, M.; Trihinas, D. Towards Efficient Resource Allocation for Federated Learning in Virtualized Managed Environments. *Future Internet* **2023**, *15*, 261. [CrossRef]

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